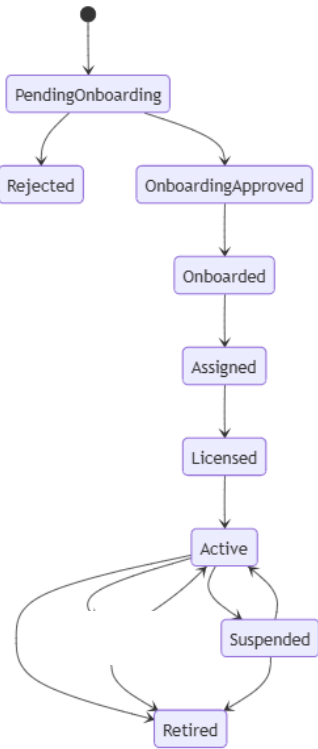


Bio-Nexa Design Principle

Bio-Nexa CMP- Device Lifecycle Model

Device Lifecycle States



State Definitions

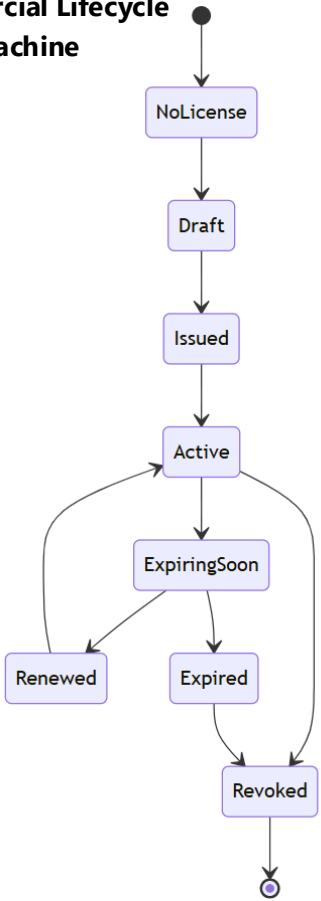
State	Description
PendingOnboarding	Device submitted onboarding request
Rejected	Onboarding request rejected
OnboardingApproved	Approved by CMP administrator
Onboarded	Device successfully provisioned and trusted
Assigned	Assigned to customer and site
Licensed	Valid license synchronized to device
Active	Device operational
Suspended	Device restricted

State Transition Matrix

From State	To State	Trigger	Performed By	Prerequisite
PendingOnboarding	OnboardingApproved	Approval	CMP Super Admin	Device validation passed
PendingOnboarding	Rejected	Reject onboarding	CMP Super Admin	Validation failed
OnboardingApproved	Onboarded	Provisioning completed	Device	Certificate installed
Onboarded	Assigned	Customer assigned	CMP Super Admin	Customer & site exist
Assigned	Licensed	License issued and synchronized	CMP Super Admin	Active license available
Licensed	Active	Device validates license locally	Device	Valid license
Active	Suspended	Administrative or policy action	Admin/System	Suspension reason
Suspended	Active	Reactivation approved	Admin	Issue resolved
Active	Retired	Retirement process	CMP Super Admin	Device decommission approved
Suspended	Retired	Permanent removal	CMP Super Admin	Retire request approved

Bio-Nexa CMP- Business and Commercial Lifecycle Model

Commercial Lifecycle State Machine



Commercial Lifecycle Flow



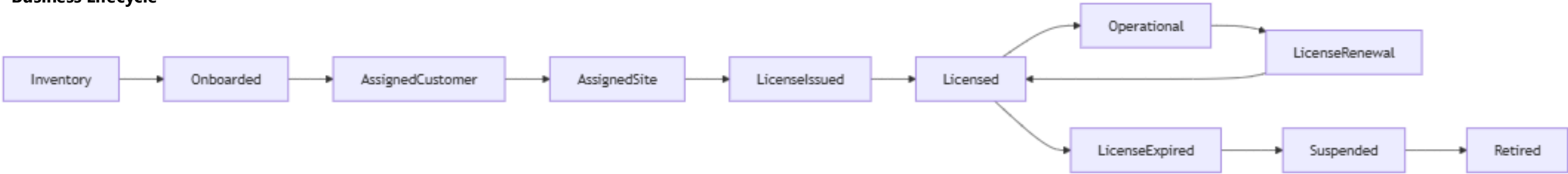
Commercial Lifecycle Flow

From	To	Trigger
NoLicense	Draft	License created
Draft	Issued	License approved
Issued	Active	License synchronized and validated
Active	ExpiringSoon	Renewal warning threshold reached
ExpiringSoon	Renewed	Renewal approved
Renewed	Active	Device receives updated license
ExpiringSoon	Expired	End date reached
Active	Revoked	Admin revocation
Expired	Revoked	Contract terminated

Commercial Business Rules

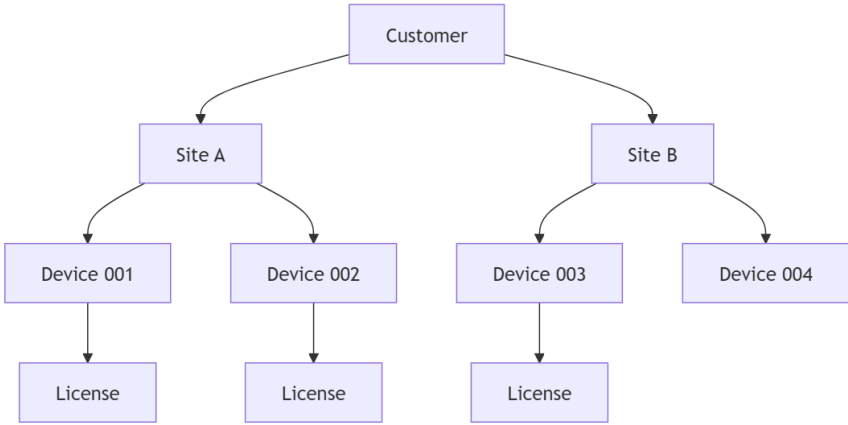
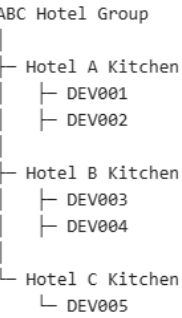
Condition	Result
Device onboarded but no customer assignment	Device not operational
Device assigned but no license	Device not operational
Device assigned and licensed	Device may enter Active state
License expired	Device enters restricted behavior according to policy

Business Lifecycle

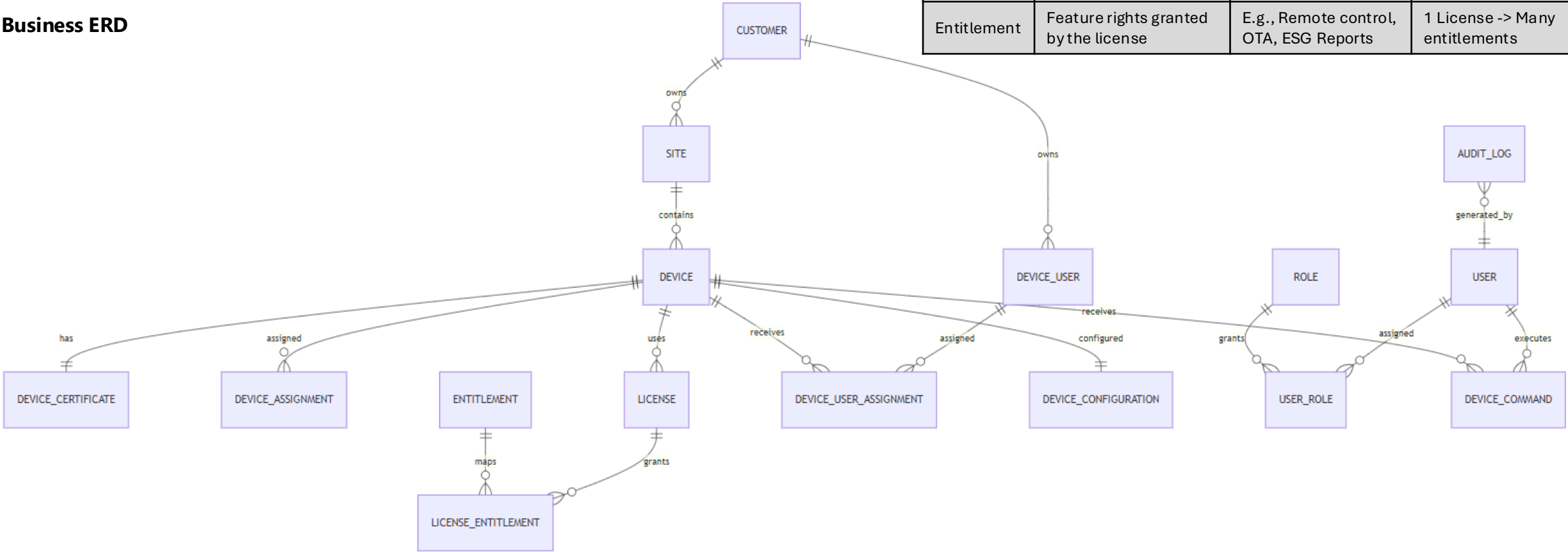


Bio-Nexa CMP- Business and Commercial Lifecycle Model

Business Hierarchy



Business ERD



Business Entity Definitions

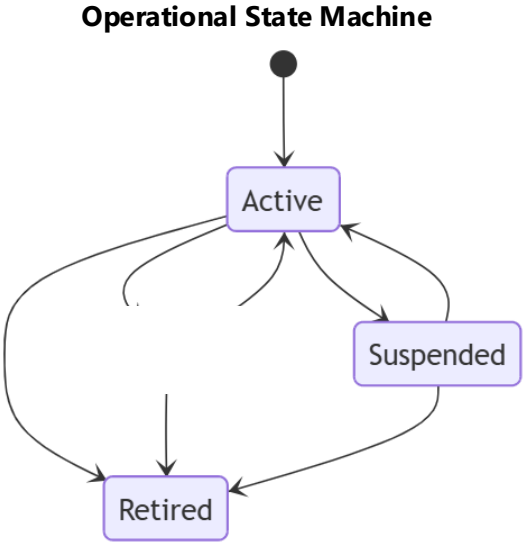
Entity	Description	Example	Relationship
Customer	Commercial owner of the deployment	ABC Hotel Group, DEF Food Court	- Own Sites - Own Devices - Own Device Users
Site	Physical Installation Location	Hotel A Kitchen, Hospital Main Kitchen	1 Customer -> Many Sites
Device	The managed Kitchen Machine	DEV001, DEV002,	1 Site -> Many Machines
License	Commercial entitlement associated with a device	Contains: license type, validity, features, status	1 Device -> 1 Active License
Entitlement	Feature rights granted by the license	E.g., Remote control, OTA, ESG Reports	1 License -> Many entitlements

Bio-Nexa CMP- Device Operational Lifecycle

Operational lifecycle controls how the device behaves

Operational State Machine

Status	Description
Active	Normal operation
Suspended	Restricted operation
Retired	Permanently decommissioned



Device Behave Matrix

Function	Active	Suspended	Retired
Local Login	✓	✓	✗
Heartbeat	✓	✓	✗
Telemetry	✓	✓	✗
Alarm Reporting	✓	✓	✗
License Sync	✓	✓	✗
Remote Control	✓	✓	✗
Device Processing	✓	✗	✗
ESG Reporting	✓	✗	✗
OTA Updates	✓	✓	✗
User Synchronization	✓	✓	✗

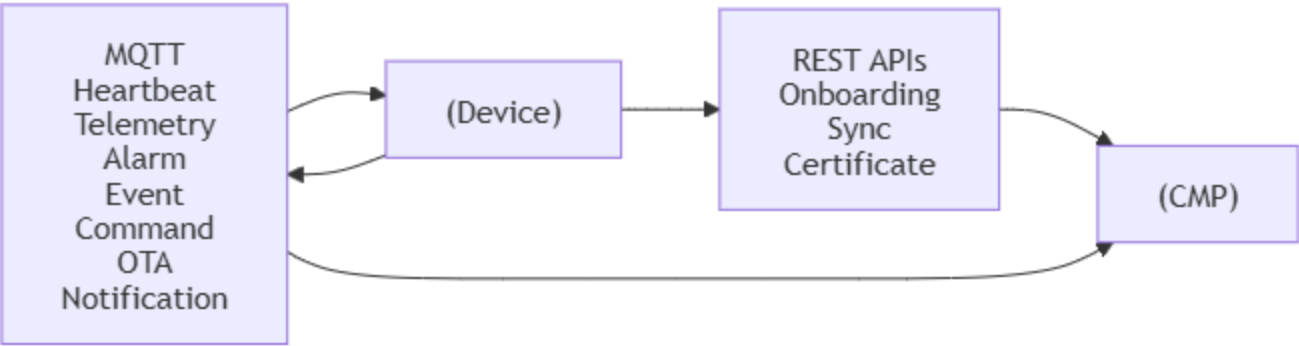
Operational Status Trigger

Event	New Status
Device commissioned	Active
License violation policy	Suspended
Security incident	Suspended
Lost/Stolen device	Suspended
End of life	Retired

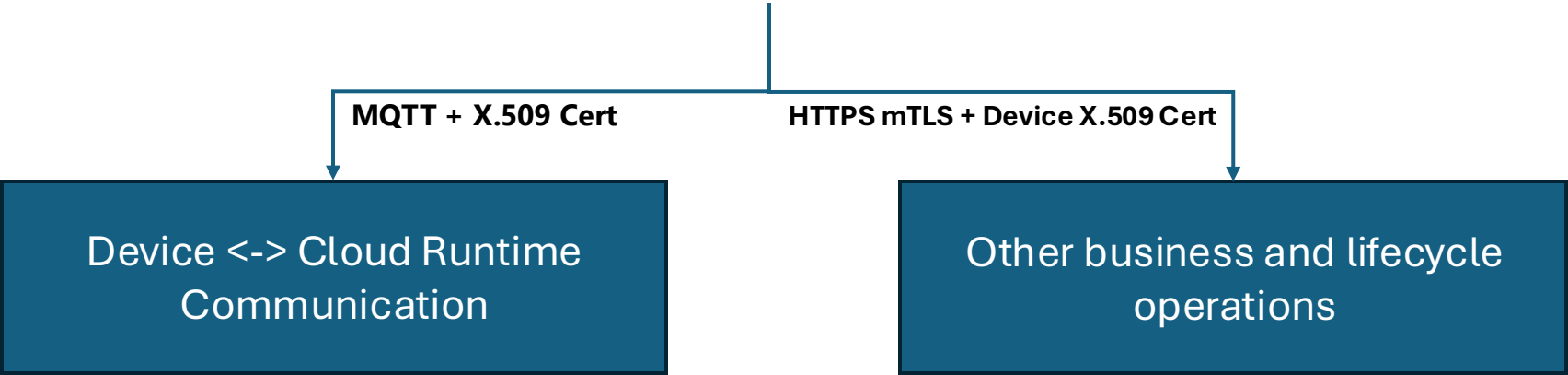
Communication Design between Device & Cloud

Bio-Nexa Cloud <-> Device Communication Principle

**Communication
Architecture**

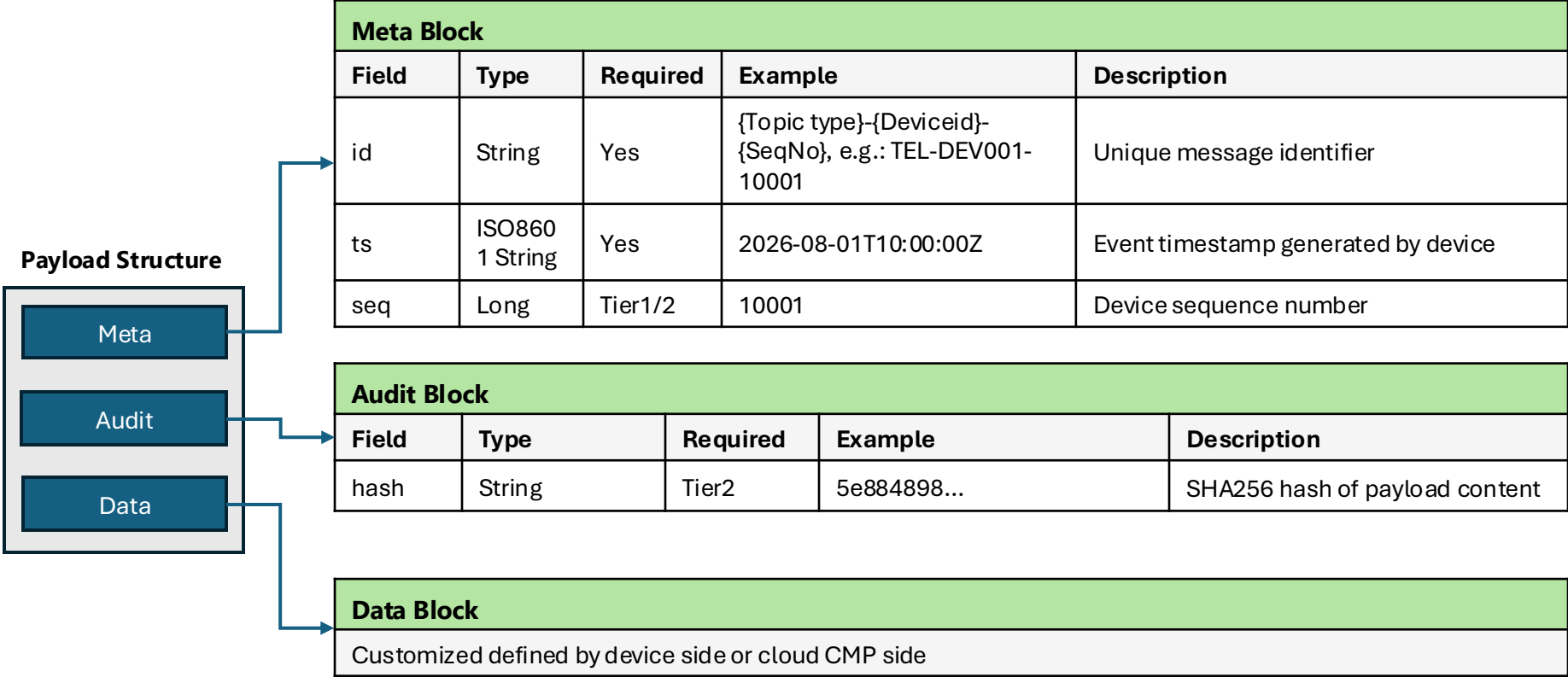


MQTT + HTTPs Restful API



Bio-Nexa MQTT Topic Design Principles – MQTT Topic Tiers and Payload Structure

Tier Definitions			
Tier	Purpose	Integrity Requirement	Payload Structure
Tier 1	Runtime Operational Data	Standard Integrity	meta + data
Tier 2	ESG / Audit Evidence	Enhanced Integrity	meta + audit + data
Tier 3	Device Control	Command Traceability	meta + data



```
{
  "meta": {
    "id": "RPT-DEV001-50001",
    "ts": "2026-08-01T23:59:59Z",
    "seq": 50001,
  },

  "audit": {
    "hash": "sha256(payload)"
  },

  "data": {
    "feedingWeight": 250,
    "dischargeWeight": 180,
    "energy": 18.5,
    "carbonReduction": 35.6
  }
}
```

Bio-Nexa MQTT Topic Catalog Design – MQTT Topics

Device → Cloud MQTT Topics

Topic Type	Purpose	Topic Path	Frequency	Contains	QoS	Tier	Payload Structure
Heartbeat	Device health and online monitoring	bnx/device/{deviceId}/heartbeat	Every 60 sec	Online status, uptime, firmware version, network status, license status	1	Tier 1	Meta + data
Telemetry	ESG sensor source telemetry data	bnx/device/{deviceId}/telemetry	10~60 sec configurable	Sensors, temperatures, humidity, gas values, power, weight	1	Tier 1	Meta + audit + data
ESG Report	ESG reporting	bnx/device/{deviceId}/report	Per cycle / hourly / daily	Feeding weight, discharge weight, energy consumption, carbon metrics	2	Tier 1	Meta + audit + data
Alarm	Fault & alert reporting	bnx/device/{deviceId}/alarm	Event driven	Alarm code, severity, details	1	Tier 1	Meta + data
Event	Operational audit events	bnx/device/{deviceId}/event	Event driven	Start, stop, mode changes, user login, reboot	1	Tier 1	Meta + data
ACK	Command execution acknowledgment	bnx/device/{deviceId}/ack	On command execution	Result, status, error code	1	Tier 1	Meta + data
Tamper (optional)	Security evidence and tamper reporting	bnx/device/{deviceId}/tamper	Event driven	Root detection, certificate mismatch, clock tampering	2	Tier 1	Meta + audit + data
Media	Camera/image upload notification	bnx/device/{deviceId}/media	Event driven	Image/video metadata, URL, timestamps	1	Tier 1	Meta + data

Cloud → Device MQTT Topics

Topic Type	Purpose	Topic Path	Frequency	Contains	QoS	Tier	Payload Structure
Command	Remote device operation	bnx/device/{deviceId}/cmd	On demand	Start, stop, forward, backward, heating, discharge, reboot	2	Tier 1	Meta + data
OTA	Softwar/firmware upgrade trigger	bnx/device/{deviceId}/ota	On demand	OTA job id, version, download URL, checksum	1	Tier 1	Meta + data
Notification	Business State change notification	bnx/device/{deviceId}/notification	On demand	SYNC_REQUIRED, LICENSE_CHANGED, CONFIG_CHANGED, STATUS_CHANGED	1	Tier 1	Meta + data

Bio-Nexa MQTT Topic Design Principles – MQTT Topic Data Sample – Heartbeat

Heartbeat Data Fields					
Category	Field	Type	Required	Example	Purpose
Device	deviceStatus	Enum	Yes	ONLINE DEGRADED OFFLINE	Device connectivity state
Device	uptimeSeconds	Long	Yes	86400	Device uptime
Device	firmwareVersion	String	Yes	1.0.5	Current software version
Operational	operationalStatus	Enum	Yes	ACTIVE SUSPENDED RETIRED	Operational lifecycle state
Operational	machineRunning	Boolean	Yes	true	Machine currently running
Operational	machineMode	Enum	Yes	IDLE PROCESSING HEATING DISCHARGING STOPPED ERROR	Current operating mode
License	licenseStatus	Enum	Yes	ACTIVE EXPIRING EXPIRED REVOKED	License state
License	licenseExpiryDate	Date	No	2027-12-31	License expiry visibility
Network	networkType	Enum	Yes	4G	Network type
Network	networkStatus	Enum	Yes	CONNECTED WEAK DISCONNECTED	Connectivity state
Network	signalStrength	Integer	No	-68	Cellular/WiFi signal
System	cpuUsagePct	Number	No	25	CPU utilization
System	memoryUsagePct	Number	No	40	Memory utilization
System	storageUsagePct	No	30	Storage utilization	
Sensors	sensorOverallStatus	Enum	Yes	NORMAL WARNING FAILED	Overall sensor health
Sensors	temperatureSensor	Enum	No	NORMAL WARNING FAILED	Individual sensor health
Sensors	humiditySensor	Enum	No	NORMAL WARNING FAILED	Individual sensor health
Sensors	weightSensor	Enum	No	NORMAL WARNING FAILED	Individual sensor health
Sensors	gasSensor	Enum	No	NORMAL WARNING FAILED	Individual sensor health
Security	certificateStatus	Enum	No	VALID EXPIRING EXPIRED REVOKED	Certificate health
Security	tamperStatus	Enum	No	NORMAL	Device integrity status



```
1 {
2   "meta": {
3     "id": "HB-DEV001-1001",
4     "ts": "2026-08-01T10:00:00Z",
5     "seq": 1001
6   },
7
8   "data": {
9
10    "deviceStatus": "ONLINE",
11
12    "operationalStatus": "ACTIVE",
13
14    "uptimeSeconds": 86400,
15
16    "licenseStatus": "ACTIVE",
17
18    "network": {
19      "type": "4G",
20      "status": "CONNECTED",
21      "signal": -68
22    },
23
24    "system": {
25      "cpuUsage": 25,
26      "memoryUsage": 40,
27      "storageUsage": 30
28    },
29
30    "machine": {
31      "running": true,
32      "currentMode": "PROCESSING"
33    },
34
35    "sensorStatus": {
36      "temperatureSensor": "NORMAL",
37      "humiditySensor": "NORMAL",
38      "weightSensor": "NORMAL",
39      "gasSensor": "NORMAL"
40    }
41  }
42 }
```

Bio-Nexa MQTT Topic Design Principles – MQTT Topic Data Sample – Telemetry

Used for:

- Real-time monitoring
- ESG calculation source data
- Trend analysis
- Data analytics

Detailed machine measurements

Category	Field	Example	Remarks
Process	feedingWeightKg	25	Current input weight
Process	chamberWeightKg	120	Current chamber weight
Process	dischargeWeightKg	5	Current discharge amount
Environment	humidityPct	45.2	Humidity reading
Environment	ambientTempC	28.5	Ambient temperature
Heat	heatTemperatureC	72.1	Heater temperature
Heat	siloTemperatureC	62.5	Silo temperature
Energy	powerConsumptionKw	2.8	Real-time power consumption
Gas	o2Pct	20.5	Oxygen concentration
Gas	co2Ppm	420	Carbon dioxide concentration
Gas	ch4Ppm	12	Methane concentration
Gas	n2oPpm	1.5	Nitrous oxide concentration
Motor	currentAmp	3.8	Motor current

```
1 {
2   "meta": {
3     "id": "TEL-DEV001-1002",
4     "ts": "2026-08-01T10:01:00Z",
5     "seq": 1002
6   },
7   "audit": {
8     "hash": "2f4d2a9e6b7c91bb"
9   },
10  "data": {
11    "feedingWeightKg": 25,
12    "chamberWeightKg": 120,
13    "dischargeWeightKg": 5,
14    "humidityPct": 45.2,
15    "ambientTempC": 28.5,
16    "heatTemperatureC": 72.1,
17    "siloTemperatureC": 62.5,
18    "powerConsumptionKw": 2.8,
19    "o2Pct": 20.5,
20    "co2Ppm": 420,
21    "ch4Ppm": 12,
22    "n2oPpm": 1.5,
23    "motorCurrentAmp": 3.8
24  }
25 }
```

Bio-Nexa MQTT Topic Design Principles – MQTT Topic Data Sample – ESG Reports

ESG Business evidence:

- Cycle summary
- Hourly summary
- Daily summary

Category	Field	Example	Remarks
Report	reportType	DAILY	CYCLE / HOURLY / DAILY
Report	periodStartTime	2026-08-01T00:00:00Z	Reporting start
Report	periodEndTime	2026-08-01T23:59:59Z	Reporting end
Throughput	feedingWeightKg	250	Total waste input
Throughput	dischargeWeightKg	180	Total waste output
Throughput	reductionWeightKg	70	Weight reduction
Processing	cycleCount	12	Completed cycles
Processing	processingDurationMinutes	480	Processing duration
Energy	energyConsumptionKwh	18.5	Total electricity consumption
Energy	averagePowerKw	2.4	Average power
Gas	averageO2Pct	20.4	Average O ₂
Gas	averageCo2Ppm	430	Average CO ₂
Gas	averageCh4Ppm	11	Average CH ₄
Gas	averageN2oPpm	1.4	Average N ₂ O
ESG	carbonReductionKg	35.6	Calculated reduction
ESG	carbonReductionMethod	DEFAULT_V1	Calculation method
Quality	dataCompletenessPct	99.8	Data quality metric
Quality	missingRecordCount	2	Missing telemetry records

```
1 {
2   "meta": {
3     "id": "RPT-DEV001-5001",
4     "ts": "2026-08-01T23:59:59Z",
5     "seq": 5001
6   },
7   "audit": {
8     "hash": "b93d177e9f4aaaf8"
9   },
10  "data": {
11    "reportType": "DAILY",
12
13    "periodStartTime": "2026-08-01T00:00:00Z",
14    "periodEndTime": "2026-08-01T23:59:59Z",
15
16    "feedingWeightKg": 250,
17    "dischargeWeightKg": 180,
18    "reductionWeightKg": 70,
19
20    "cycleCount": 12,
21    "processingDurationMinutes": 480,
22
23    "energyConsumptionKwh": 18.5,
24    "averagePowerKw": 2.4,
25
26    "averageO2Pct": 20.4,
27    "averageCo2Ppm": 430,
28    "averageCh4Ppm": 11,
29    "averageN2oPpm": 1.4,
30
31    "carbonReductionKg": 35.6,
32    "carbonReductionMethod": "DEFAULT_V1",
33
34    "dataCompletenessPct": 99.8,
35    "missingRecordCount": 2
36  }
37 }
```

Bio-Nexa MQTT Topic Design Principles – MQTT Topic Data Sample – Alarm

Purpose

Alarm should be an actionable exception requiring attention.

Not a sensor reading.

Used for:

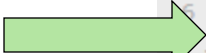
- Device Faults
- Maintenance Actions
- Safety Events
- Escalation

Category	Field	Example	Remarks
Alarm	code	TEMP_HIGH	Alarm code
Alarm	category	TEMPERATURE	Alarm category
Alarm	severity	HIGH	INFO/WARNING/HIGH/CRITICAL
Alarm	status	ACTIVE	ACTIVE/CLEARED
Alarm	detectedTime	2026-08-01T10:15:00Z	Detection time
Alarm	component	HEATER	Faulting component
Alarm	currentValue	88	Current value
Alarm	threshold	80	Threshold value
Alarm	unit	C	Measurement unit
Alarm	message	Heating temperature exceeded threshold	Human-readable message
Alarm	recommendedAction	CHECK_HEATER	Maintenance recommendation

```
1 {
2   "meta": {
3     "id": "ALM-DEV001-3001",
4     "ts": "2026-08-01T10:15:00Z",
5     "seq": 3001
6   },
7   "data": {
8     "code": "TEMP_HIGH",
9     "category": "TEMPERATURE",
10
11     "severity": "HIGH",
12     "status": "ACTIVE",
13
14     "detectedTime": "2026-08-01T10:15:00Z",
15
16     "component": "HEATER",
17
18     "currentValue": 88,
19     "threshold": 80,
20     "unit": "C",
21
22     "message": "Heating temperature exceeded threshold",
23
24     "recommendedAction": "CHECK_HEATER"
25   }
26 }
```

Bio-Nexa MQTT Topic Design Principles – MQTT Topic Data Sample – Event

Category	Field	Example	Remarks
Event	eventType	MACHINE_STARTED	Event type
Event	userId	USR001	Operator ID
Event	username	operator01	Operator account
Event	source	LOCAL	LOCAL / REMOTE
Event	remarks	Daily operation start	Free text remarks



```
1 {
2   "meta": {
3     "id": "EVT-DEV001-4001",
4     "ts": "2026-08-01T10:30:00Z",
5     "seq": 4001
6   },
7   "data": {
8     "eventType": "MACHINE_STARTED",
9     "userId": "USR001",
10    "username": "operator01",
11    "source": "LOCAL",
12    "remarks": "Daily production operation started"
13  }
14 }
```

Bio-Nexa MQTT Topic Design Principles – MQTT Topic Data Sample – ACK

Category	Field	Example	Remarks
Command	commandId	CMD-DEV001-6001	Related command
Command	command	START	Executed command
Result	result	SUCCESS	SUCCESS / FAILED
Result	executeTimeMs	523	Processing time
Result	errorCode	null	Error code when failed
Result	message	Machine Started	Execution result



```
1 {
2   "meta": {
3     "id": "ACK-DEV001-6001",
4     "ts": "2026-08-01T10:35:00Z",
5     "seq": 6001
6   },
7   "data": {
8     "commandId": "CMD001",
9     "command": "START",
10    "result": "SUCCESS",
11    "executeTimeMs": 523,
12    "errorCode": null,
13    "message": "Machine started successfully"
14  }
15 }
```

Bio-Nexa MQTT Topic Design Principles – MQTT Topic Data Sample – Tamper

Category	Field	Example	Remarks
Security	eventType	ROOT_DETECTED	Security event
Security	severity	CRITICAL	Security severity
Security	component	ANDROID_OS	Affected component
Security	details	Root binary detected	Detailed description
Security	actionTaken	DEVICE_SUSPENDED	Automatic response



```
1 {
2   "meta": {
3     "id": "TMP-DEV001-7001",
4     "ts": "2026-08-01T12:00:00Z",
5     "seq": 7001
6   },
7   "audit": {
8     "hash": "d3f992ab371cd221"
9   },
10  "data": {
11    "eventType": "ROOT_DETECTED",
12    "severity": "CRITICAL",
13    "component": "ANDROID_OS",
14    "details": "Root binary detected",
15    "actionTaken": "DEVICE_SUSPENDED"
16  }
17 }
```

Bio-Nexa MQTT Topic Design Principles – MQTT Topic Data Sample – Media

Category	Field	Example	Remarks
Media	mediaType	IMAGE	IMAGE / VIDEO
Media	captureTime	2026-08-01T12:04:55Z	Original capture time
Media	fileName	snapshot.jpg	File name
Media	objectPath	s3://bucket/path	Cloud storage path
Media	sizeKb	512	File size
Media	durationSec	0	Video duration, 0 for image



```
1 {
2   "meta": {
3     "id": "MED-DEV001-8001",
4     "ts": "2026-08-01T12:05:00Z",
5     "seq": 8001
6   },
7   "data": {
8     "mediaType": "IMAGE",
9     "captureTime": "2026-08-01T12:04:55Z",
10    "fileName": "snapshot_20260801_120455.jpg",
11    "objectPath": "s3://bnx-media/dev001/snapshot_20260801_120455.jpg",
12    "sizeKb": 512,
13    "durationSec": 0
14  }
15 }
```

Bio-Nexa MQTT Topic Design Principles – MQTT Topic Data Sample – Command

Category	Field	Example	Remarks
Command	command	START	Command name
Command	requestedBy	tenant_admin	Request initiator
Command	requestTime	2026-08-01T10:00:00Z	Original request time
Command	timeoutSec	30	Execution timeout
Command	remarks	Start processing batch	Optional note



```
1 {
2   "meta": {
3     "id": "CMD-DEV001-001",
4     "ts": "2026-08-01T10:00:00Z"
5   },
6   "data": {
7     "command": "START",
8     "requestedBy": "tenant_admin",
9     "requestTime": "2026-08-01T10:00:00Z",
10    "timeoutSec": 30,
11    "remarks": "Start processing operation"
12  }
13 }
```

Machine Operation Commands

Command	Description
START	Start processing cycle
STOP	Stop processing cycle
PAUSE	Pause processing
RESUME	Resume processing
EMERGENCY_STOP	Immediate stop

Heating Commands

Command	Description
HEATING_ON	Enable heating
HEATING_OFF	Disable heating
SET_TARGET_TEMPERATURE	Set target temperature

Discharge Commands

Command	Description
DISCHARGE_START	Start discharge
DISCHARGE_STOP	Stop discharge

Motor Commands

Command	Description
AGITATOR_FORWARD	Forward rotation
AGITATOR_REVERSE	Reverse rotation
AGITATOR_STOP	Stop agitator

Ventilation Commands

Command	Description
EXHAUST_ON	Start exhaust
EXHAUST_OFF	Stop exhaust
AIR_SUPPLY_ON	Start air supply
AIR_SUPPLY_OFF	Stop air supply

Device Commands

Command	Description
REBOOT	Restart device
SHUTDOWN	Shutdown device
FACTORY_RESET	Restore factory settings
TAKE_SNAPSHOT	Capture image
FORCE_SYNC	Trigger immediate sync

Bio-Nexa MQTT Topic Design Principles – MQTT Topic Data Sample – OTA

Category	Field	Example	Remarks
OTA	version	2.1.0	Target version
OTA	packageType	APP	APP / FIRMWARE
OTA	downloadUrl	https://...	Package URL
OTA	sha256	ABC123...	Package checksum
OTA	mandatory	false	Mandatory update flag
OTA	scheduledTime	2026-08-01T23:00:00Z	Planned execution time

```
1 {
2   "meta": {
3     "id": "OTA-DEV001-001",
4     "ts": "2026-08-01T11:00:00Z"
5   },
6   "data": {
7     "version": "2.1.0",
8     "packageType": "APP",
9     "downloadUrl": "https://download.bionexa.com/ota/app-v2.1.0.apk",
10    "sha256": "ABCD1234EFGH5678",
11    "mandatory": false,
12    "scheduledTime": "2026-08-01T23:00:00Z"
13  }
14 }
```

Bio-Nexa MQTT Topic Design Principles – MQTT Topic Data Sample – Notification

Category	Field	Example	Remarks
Notification	type	LICENSE_CHANGED	Notification type
Notification	priority	NORMAL	LOW/NORMAL/HIGH
Notification	title	License Updated	Display title
Notification	message	New license is available	Message content
Notification	action	SYNC	Recommended device action



```
1 {
2   "meta": {
3     "id": "NOT-DEV001-001",
4     "ts": "2026-08-01T10:45:00Z"
5   },
6   "data": {
7     "type": "LICENSE_CHANGED",
8     "priority": "NORMAL",
9     "title": "License Updated",
10    "message": "A new license is available.",
11    "action": "SYNC"
12  }
13 }
```

Synchronization Notifications

Type	Device Action
SYNC_REQUIRED	Call /api/v1/device/sync
LICENSE_CHANGED	Call /api/v1/device/sync
CONFIG_CHANGED	Call /api/v1/device/sync
USERS_CHANGED	Call /api/v1/device/sync
STATUS_CHANGED	Call /api/v1/device/sync
ASSIGNMENT_CHANGED	Call /api/v1/device/sync

OTA Notifications

Type	Device Action
OTA_AVAILABLE	Expect OTA topic
OTA_CANCELLED	Cancel pending upgrade

Certificate Notifications

Type	Device Action
CERTIFICATE_EXPIRING	Check certificate status
CERTIFICATE_ROTATION_REQUIRED	Call certificate rotation API

Security Notifications

Type	Device Action
SECURITY_POLICY_UPDATED	Sync latest policy
DEVICE_SUSPENDED	Enter suspended mode
DEVICE_RETIRED	Enter retired mode

Bio-Nexa MQTT Topic Catalog Design – Topic Data Retention Policy

Topic	Store in RDS	Store in S3 Archive
Heartbeat	Latest State Only	No
Telemetry	Aggregated Summary	Yes
ESG Report	Yes	Yes
Alarm	Yes	Yes
Event	Yes	Yes
ACK	Yes	Optional
License	Yes	Optional
Tamper	Yes	Yes
OTA	Yes	Optional

Bio-Nexa Cloud <-> Device Restful API

All APIs below are:	
Caller	Device (Android Edge Gateway)
Server	CMP
Protocol	HTTPS
Authentication (Pre-Onboarding)	QR/Onboarding Token
Authentication (Post-Onboarding)	X.509 + mTLS
Common API Path Prefix	/api/v1/
Purpose	Device Lifecycle, License, User, Configuration & Status Synchronization

Category	API	Method	Authentication	Purpose
Onboarding	/api/v1/device/onboarding/request	POST	Onboarding Token	Submit onboarding request
Onboarding	/api/v1/device/onboarding/status	GET	Onboarding Token	Retrieve onboarding approval status
Certificate	/api/v1/device/certificate/rotate	POST	X.509 + mTLS	Retrieve replacement certificate
Certificate	/api/v1/device/certificate/status	GET	X.509 + mTLS	Retrieve certificate validity information
Synchronization	/api/v1/device/sync	POST	X.509 + mTLS	Synchronize device state from CMP
Device Lifecycle	/api/v1/device/deactivate	POST	X.509 + mTLS	Device deactivation confirmation

Device Lifecycle State	API Used	Purpose
Pending Onboarding	onboarding/request	Submit onboarding request
Pending Approval	onboarding/status	Poll approval status
Approved	onboarding/status	Retrieve certificate and initial configuration
Onboarded	MQTT Heartbeat	First heartbeat proves successful onboarding
Assigned	sync	Retrieve assignment
Licensed	sync	Retrieve license
Active	sync + MQTT	Runtime operation
Maintenance	sync	Retrieve operational policy
Suspended	sync	Retrieve operational policy
Certificate Renewal	certificate/status + certificate/rotate	Certificate lifecycle
Retired	sync	Device receives retired status

Recommended Call Frequency	
API	Frequency
onboarding/request	One time
Onboarding/status	Every 30 seconds until approved/rejected
sync	See on sync api page
certificate/status	Daily
certificate/rotate	Only when instructed or nearing expiry

Bio-Nexa Cloud <-> Device Restful API – Onboarding APIs

Submit On-boarding Request

POST /api/v1/device/onboarding/request

Purpose: Initiate onboarding

Request

```
{
  "serialNumber": "BNX2026000001",
  "model": "BNX-100",
  "hardwareVersion": "1.0",
  "manufacturer": "Bio-Nexa",
  "manufactureDate": "2026-07-01"
}
```

Response

```
{
  "requestId": "REQ001",
  "status": "PENDING"
}
```

Query Onboarding Status

GET /api/v1/device/onboarding/status

Purpose: Check approval status

Response

```
{
  "status": "PENDING"
}

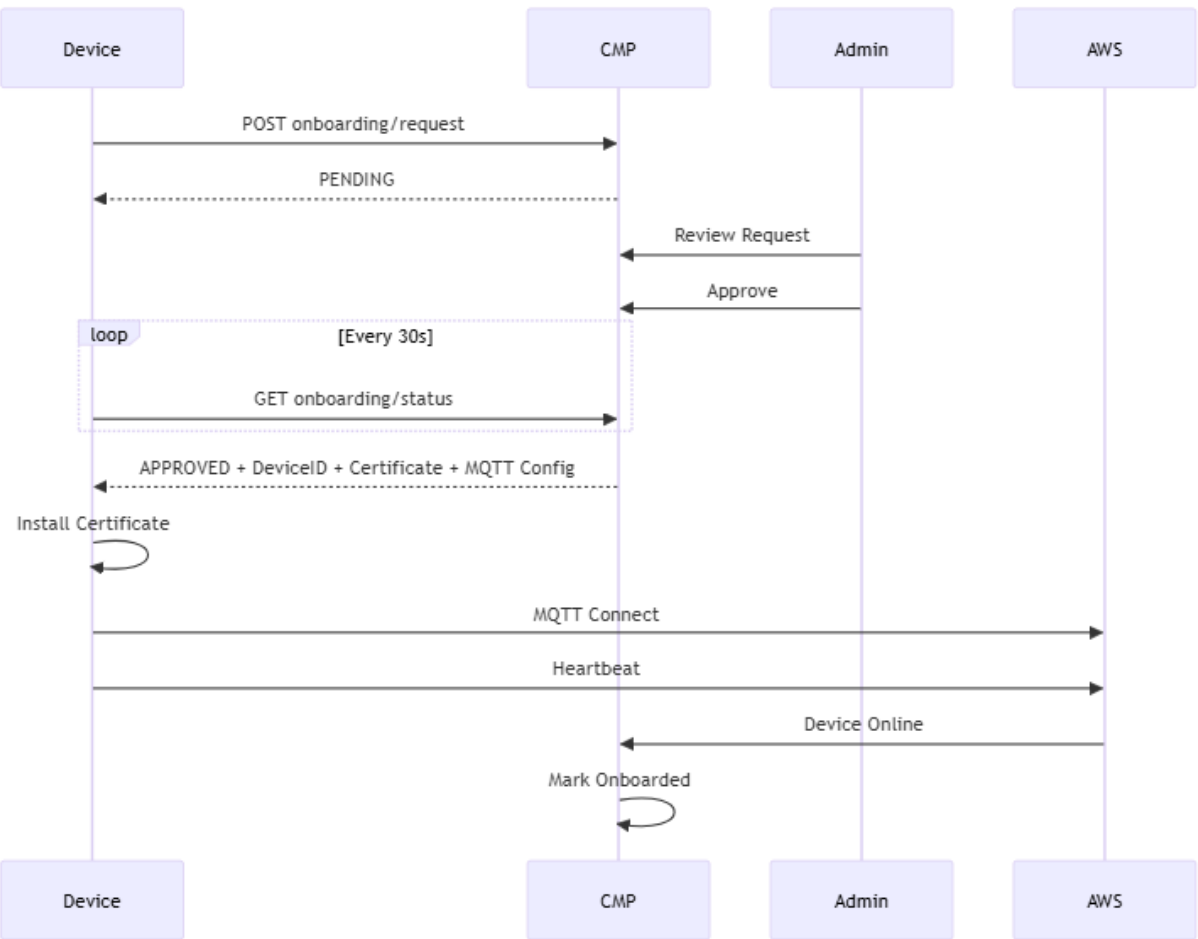
{
  "status": "REJECTED",
  "reason": "Serial Number Not Authorized"
}

{
  "status": "APPROVED",
  "deviceId": "DEV001",

  "certificate": {
    "certificatePem": "...",
    "privateKey": "..."
  },

  "mqtt": {
    "endpoint": "xxxxx.iot.ap-southeast-1.amazonaws.com"
  },

  "configuration": {
    "heartbeatInterval": 60
  }
}
```



Bio-Nexa Cloud <-> Device Restful API – Certificate APIs

Certificate Rotation

POST /api/v1/device/certificate/rotate

Purpose: Retrieve new certs before expiry

Request

```
{
  "currentCertificateId": "CERT001"
}
```

Response

```
{
  "certificateId": "CERT002",
  "certificatePem": "...",
  "privateKey": "...",
  "effectiveDate": "2027-01-01",
  "expiryDate": "2028-01-01"
}
```

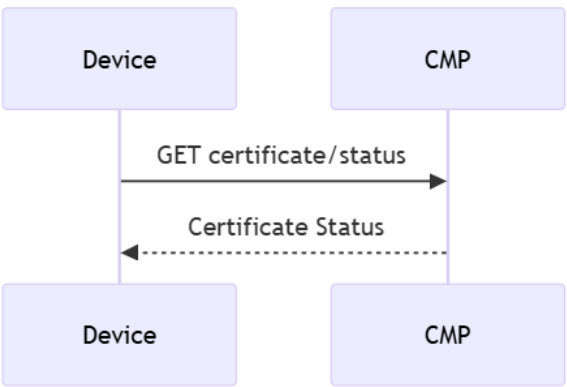
Certificate Status

GET /api/v1/device/certificate/status

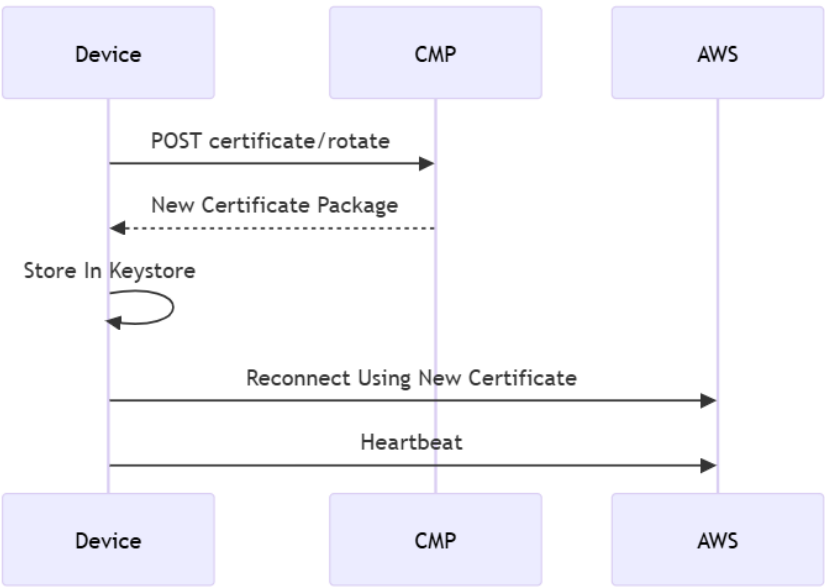
Response

```
{
  "certificateId": "CERT002",
  "status": "ACTIVE",
  "expiryDate": "2028-01-01",
  "daysRemaining": 365
}
```

Certificate Status Check



Certificate Rotation



Bio-Nexa Cloud <-> Device Restful API – Unified Device Synchronization

Unified Device Synchronization

POST /api/v1/device/sync

Purpose: Single source of truth
synchronization for:

- Customer Assignment
- Site Assignment
- License
- Entitlements
- Operational Status
- Configuration
- Device Users
- Device Alias

Request

```
{
  "lastSyncTime": "2026-08-01T10:00:00Z"
}
```

Response

```
{
  "assignment": {
    "customerId": "CUS001",
    "customerName": "ABC Hotel",

    "siteId": "SITE001",
    "siteName": "Hotel A Kitchen"
  },

  "device": {
    "alias": "Kitchen Processor 01"
  },

  "license": {
    "licenseId": "LIC001",

    "status": "ACTIVE",

    "validFrom": "2026-01-01",

    "validTo": "2027-12-31",

    "entitlements": [
      "REMOTE_CONTROL",
      "OTA",
      "ESG_REPORTING"
    ],

    "signature": "..."
  },

  "configuration": {
    "heartbeatInterval": 60,

    "telemetryInterval": 30,

    "cameraRefreshInterval": 1,

    "temperatureThreshold": 80
  },

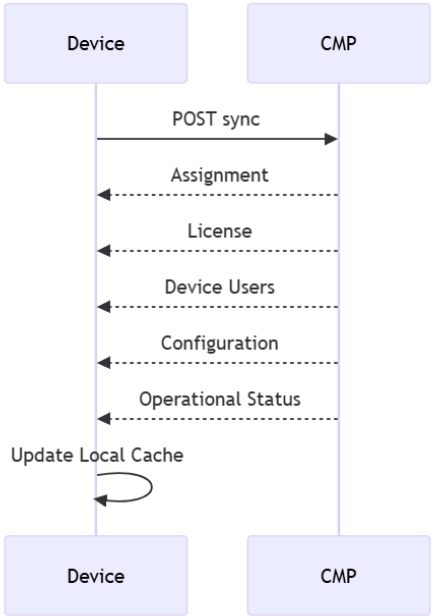
  "deviceUsers": [
    {
      "userId": "USR001",

      "username": "operator01",

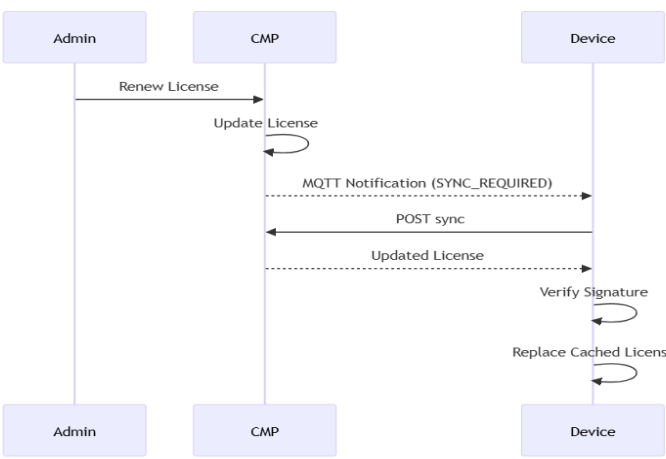
      "displayName": "Kitchen Operator",

      "passwordHash": "...",

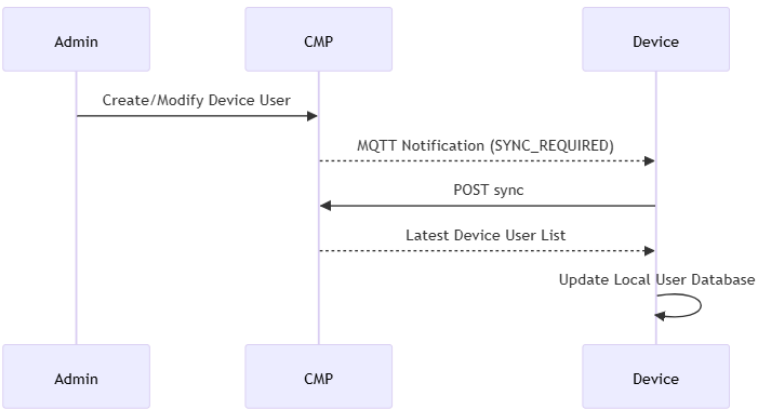
      "status": "ACTIVE"
    }
  ]
}
```



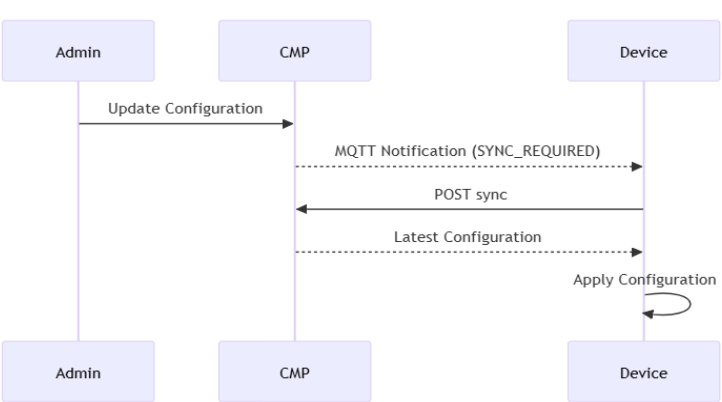
License Renewal Flow



Device User Synchronization

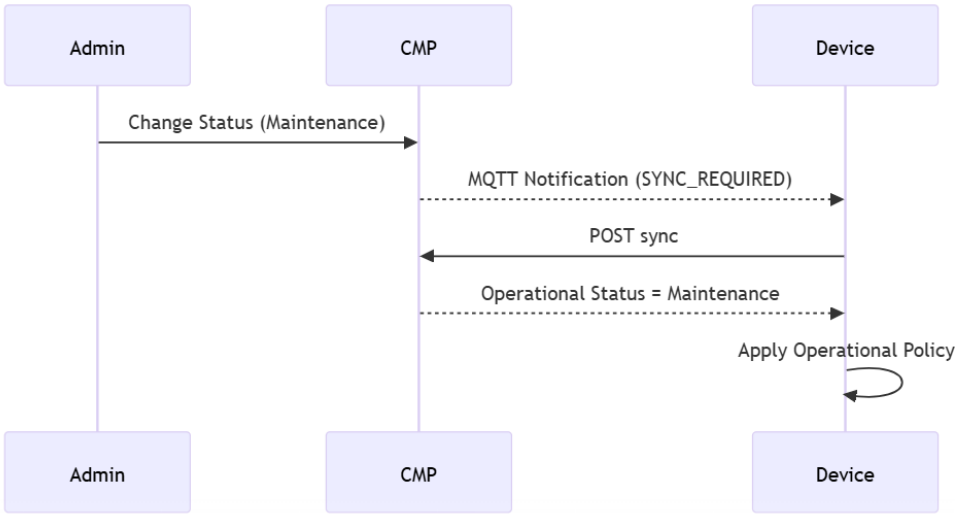


Configuration Synchronization

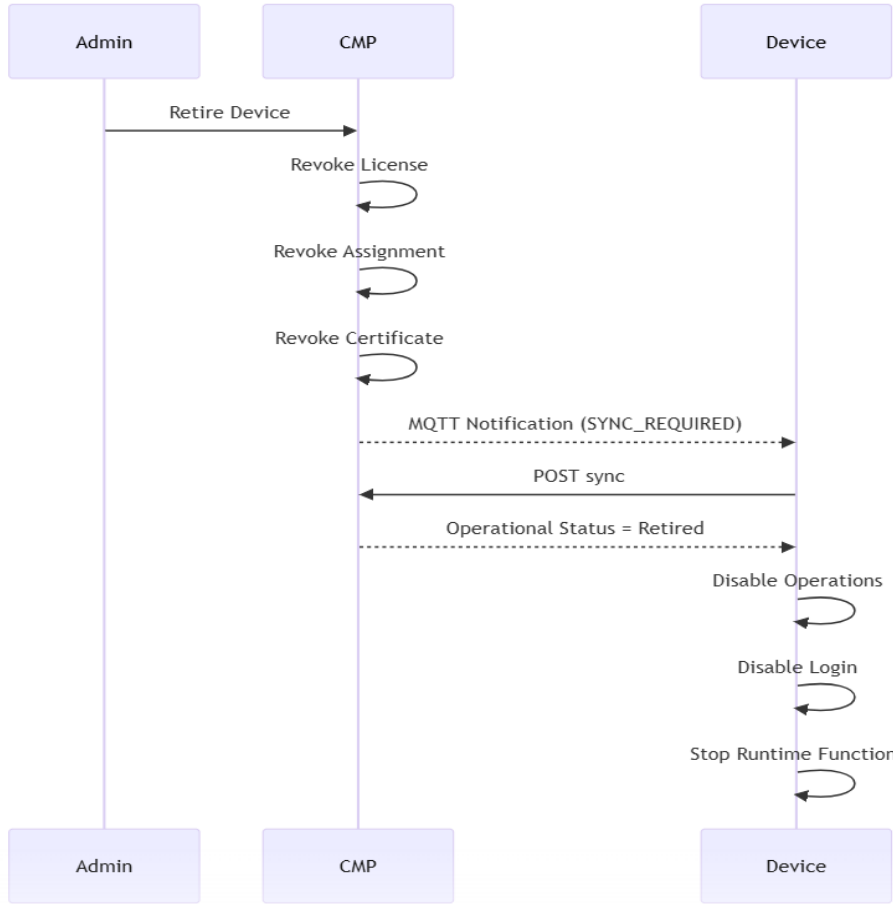


Bio-Nexa Cloud <-> Device Restful API – Unified Device Synchronization

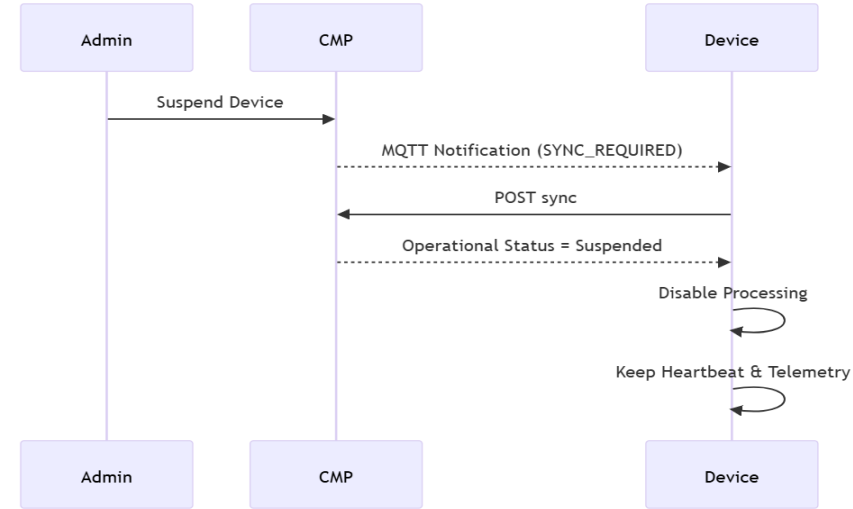
Operational Status Change Synchronization



Device Retirement



Device Suspension



Runtime Trigger Pattern



Bio-Nexa Cloud <-> Device Restful API – Unified Device Synchronization

Synchronization Contents

Domain	Description
Assignment	Customer, Site, Authorization Window
Device Metadata	Alias
License	Status, validity, entitlements
Device Users	Latest authorized device operators
Configuration	Latest active parameters
Operational Status	Active / Maintenance / Suspended / Retired

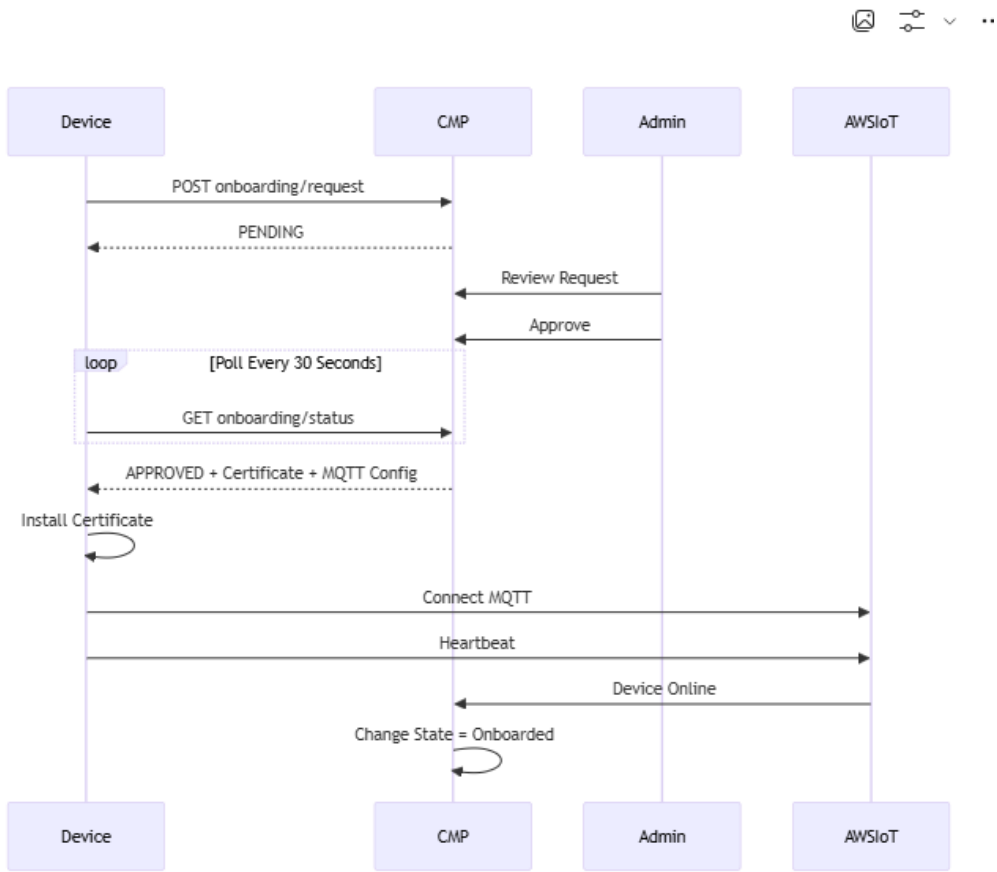
Sync API Calling Frequency

Device Status	Frequency
Active	Every 5 minutes
Suspended	Every 15 minutes
Offline Recovery	Immediately after reconnect
MQTT Sync Notification	Immediately

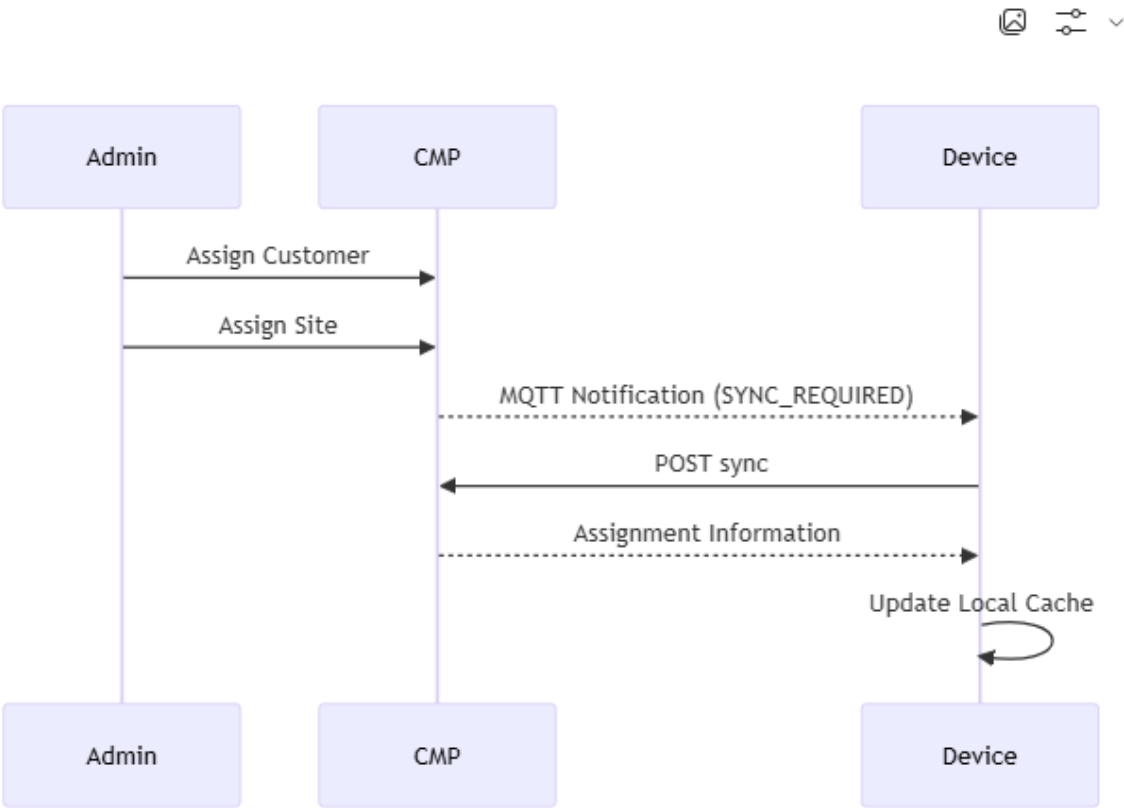
Business Scenarios System Integration Flow

Scenarios

1. Device Onboarding

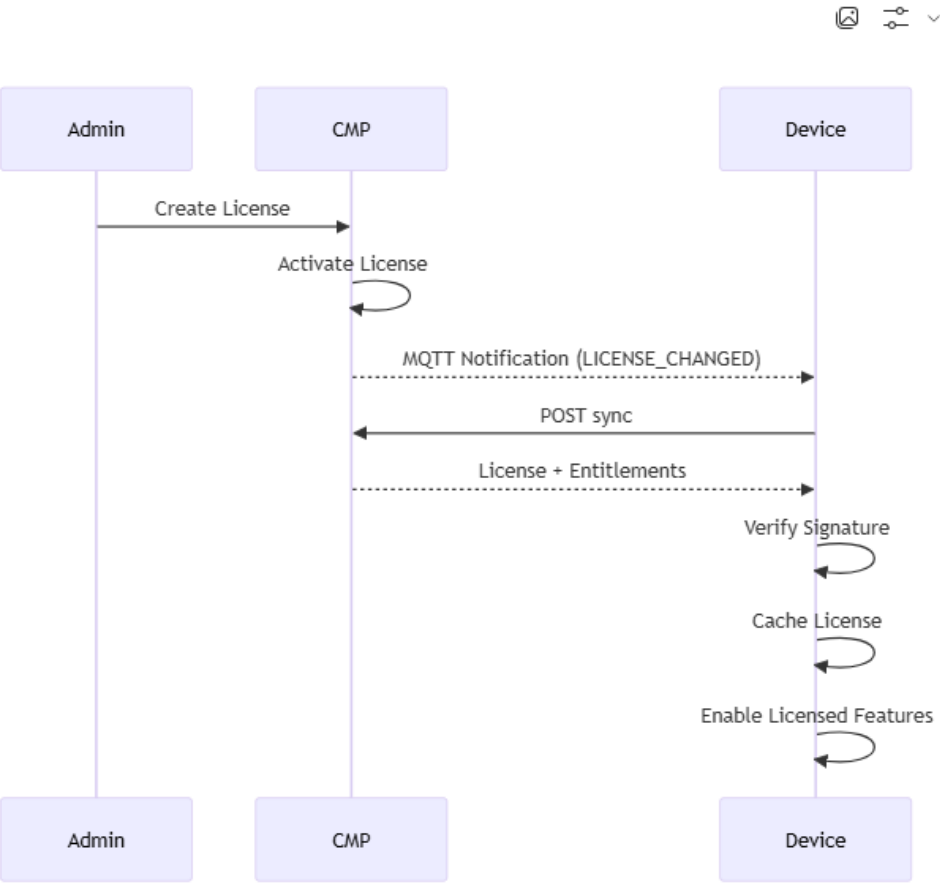


2. First Device Assignment

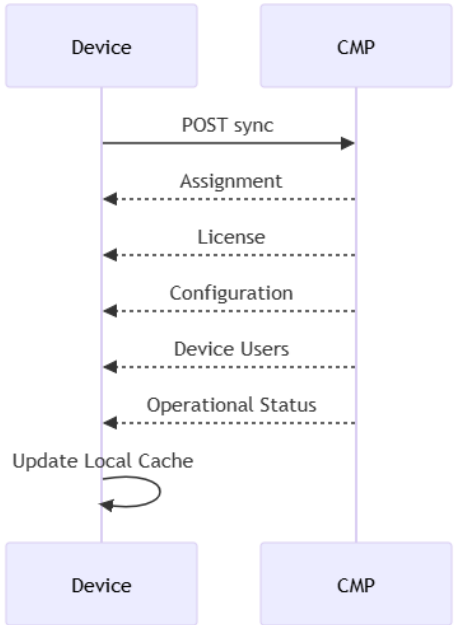


Scenarios

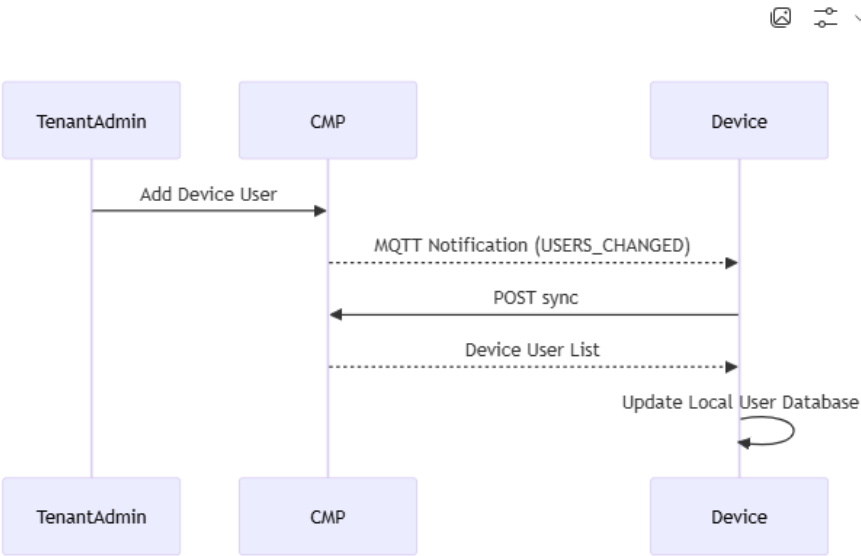
3. License Creation & Distribution



4. Device Periodic Synchronization

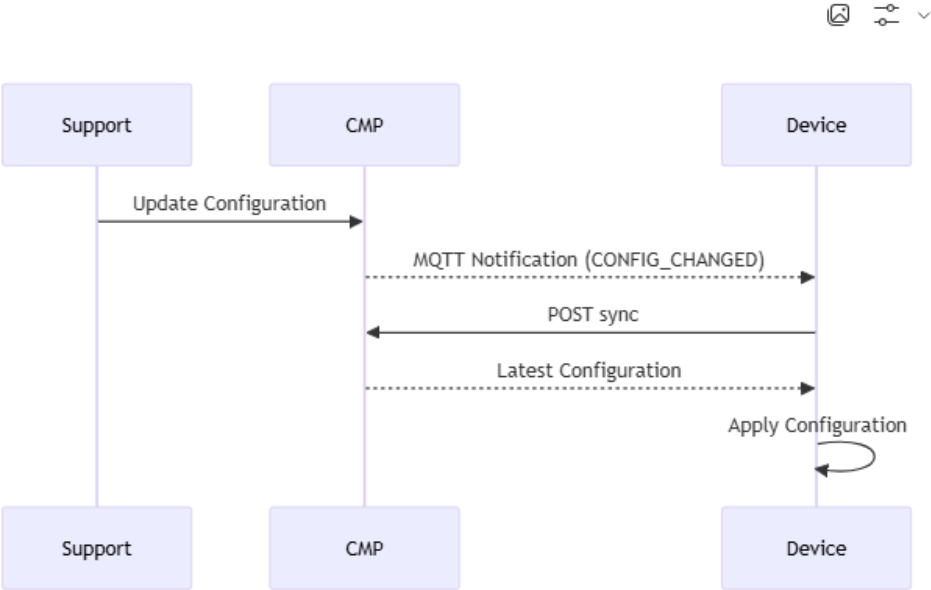


5. Device User Synchronization

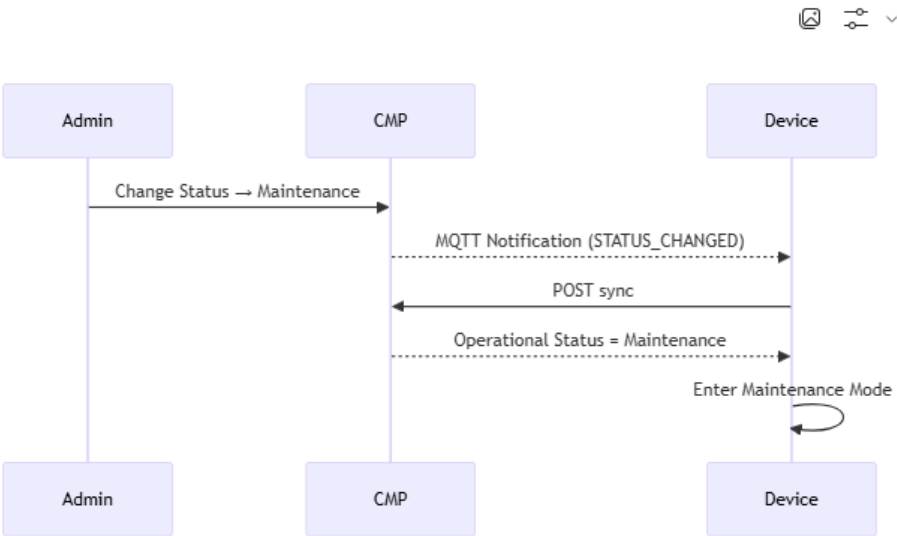


Scenarios

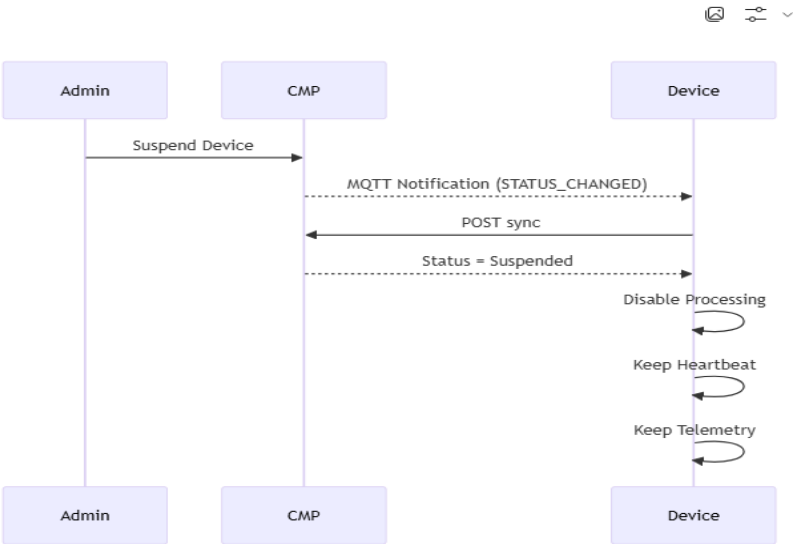
6. Configuration Change



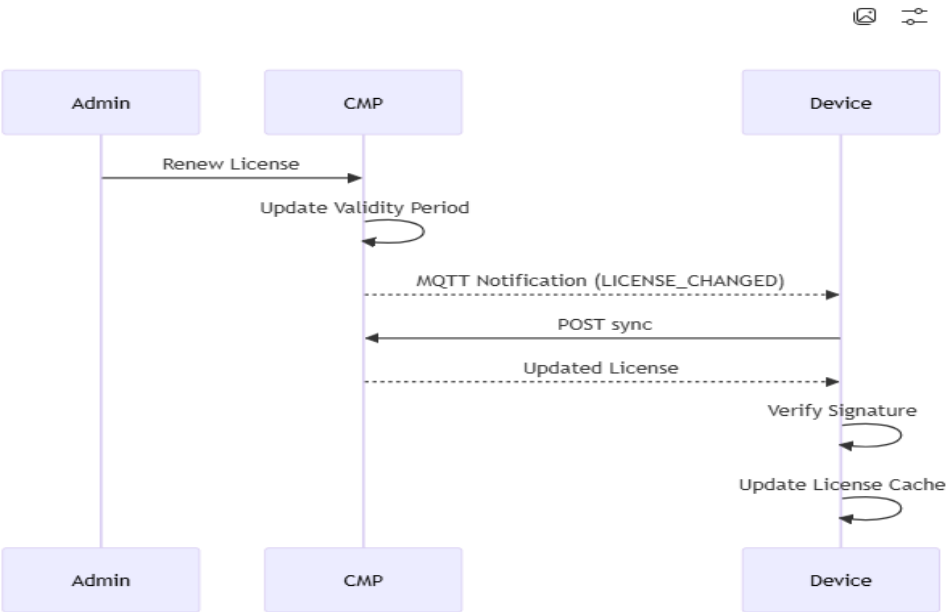
7. Operational Status Change



8. Device Suspension

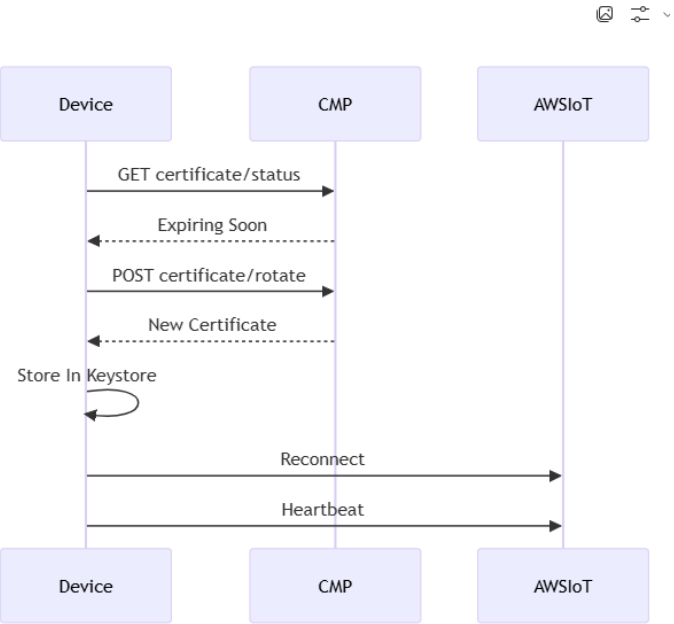


9. License Renewal

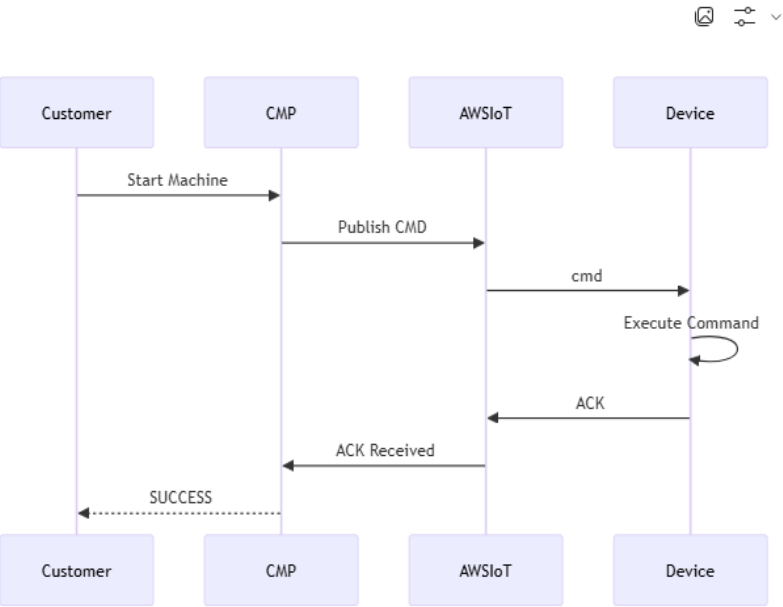


Scenarios

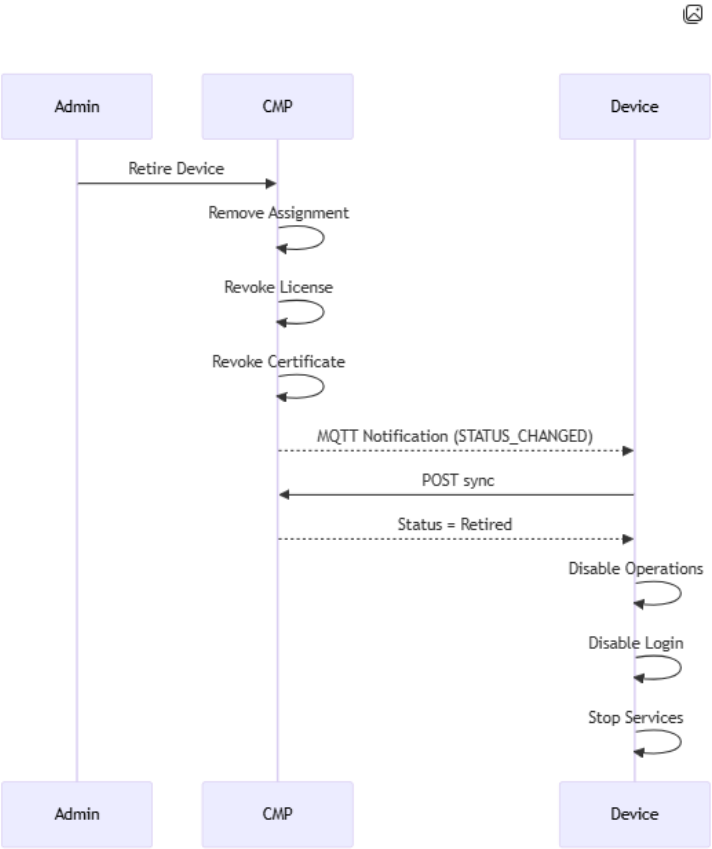
10. Certificate Rotation



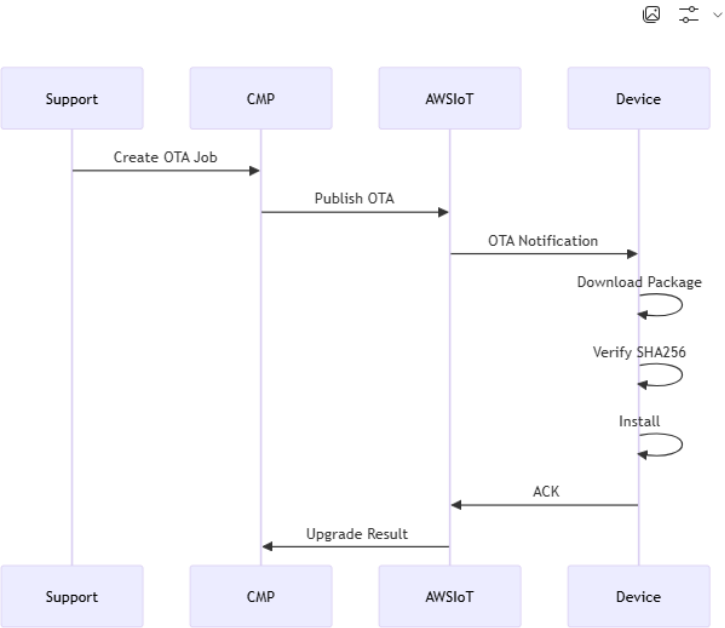
11. Remote Command Execution



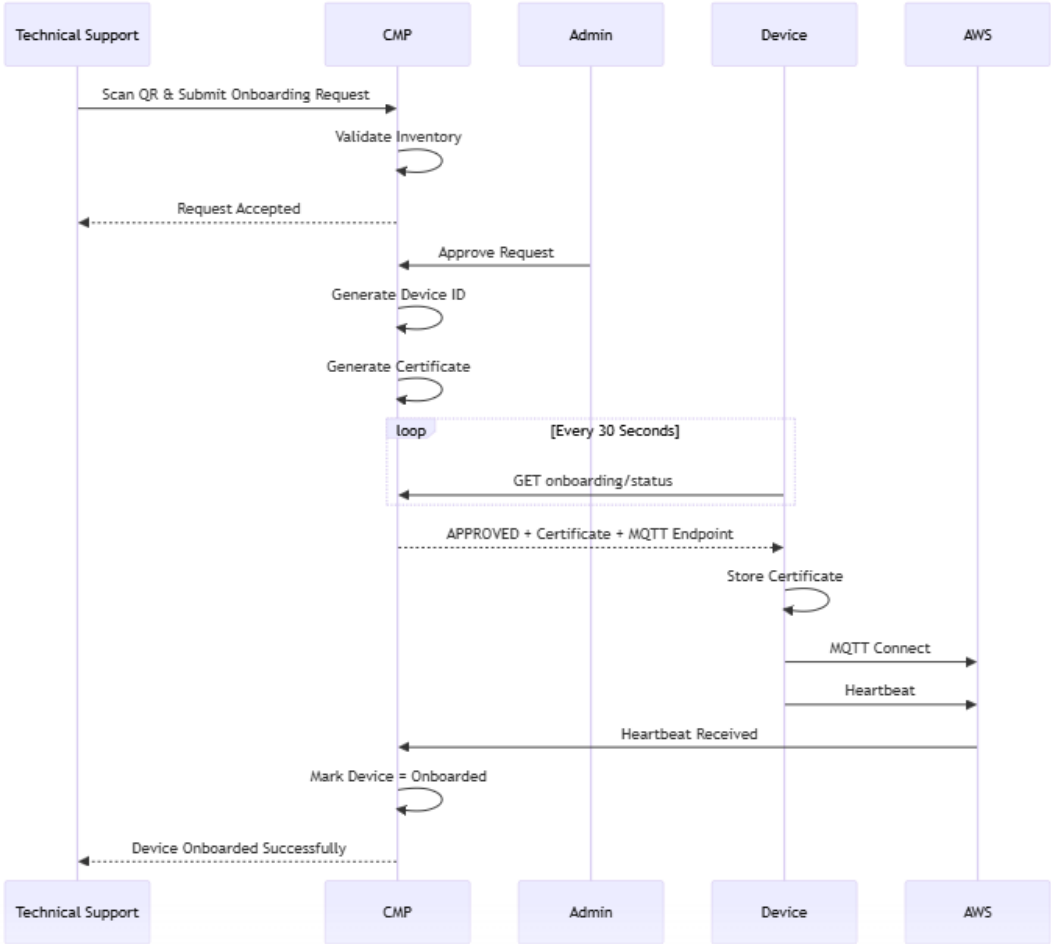
13. Device Retirement



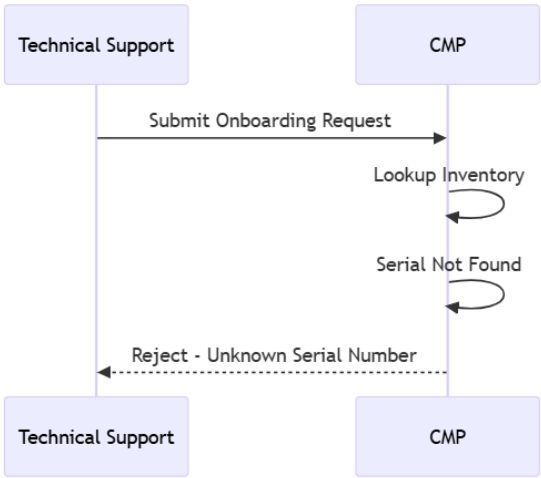
12. OTA Upgrade



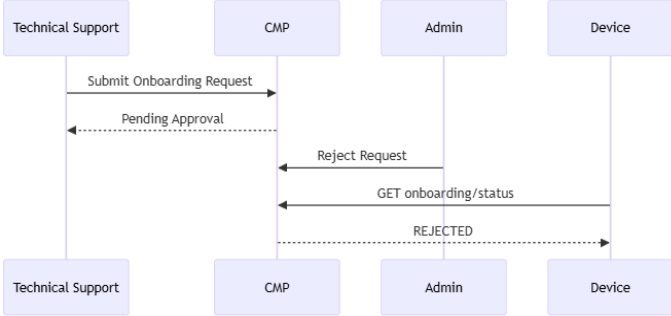
Onboarding Process More details with negative and positive cases



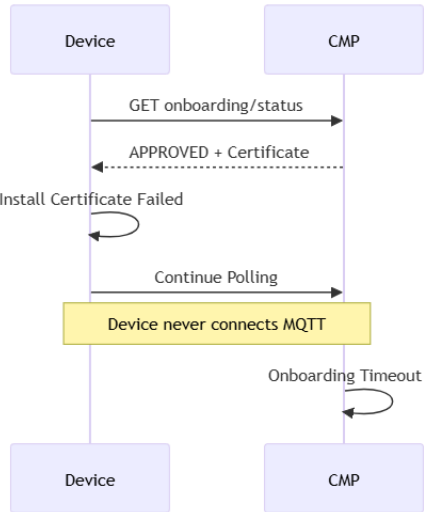
Negative Case 1: Serial Number Not Found



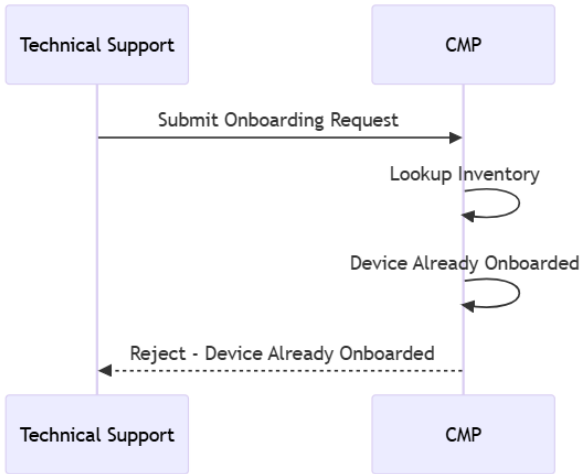
Negative Case 3: Admin Rejects Request



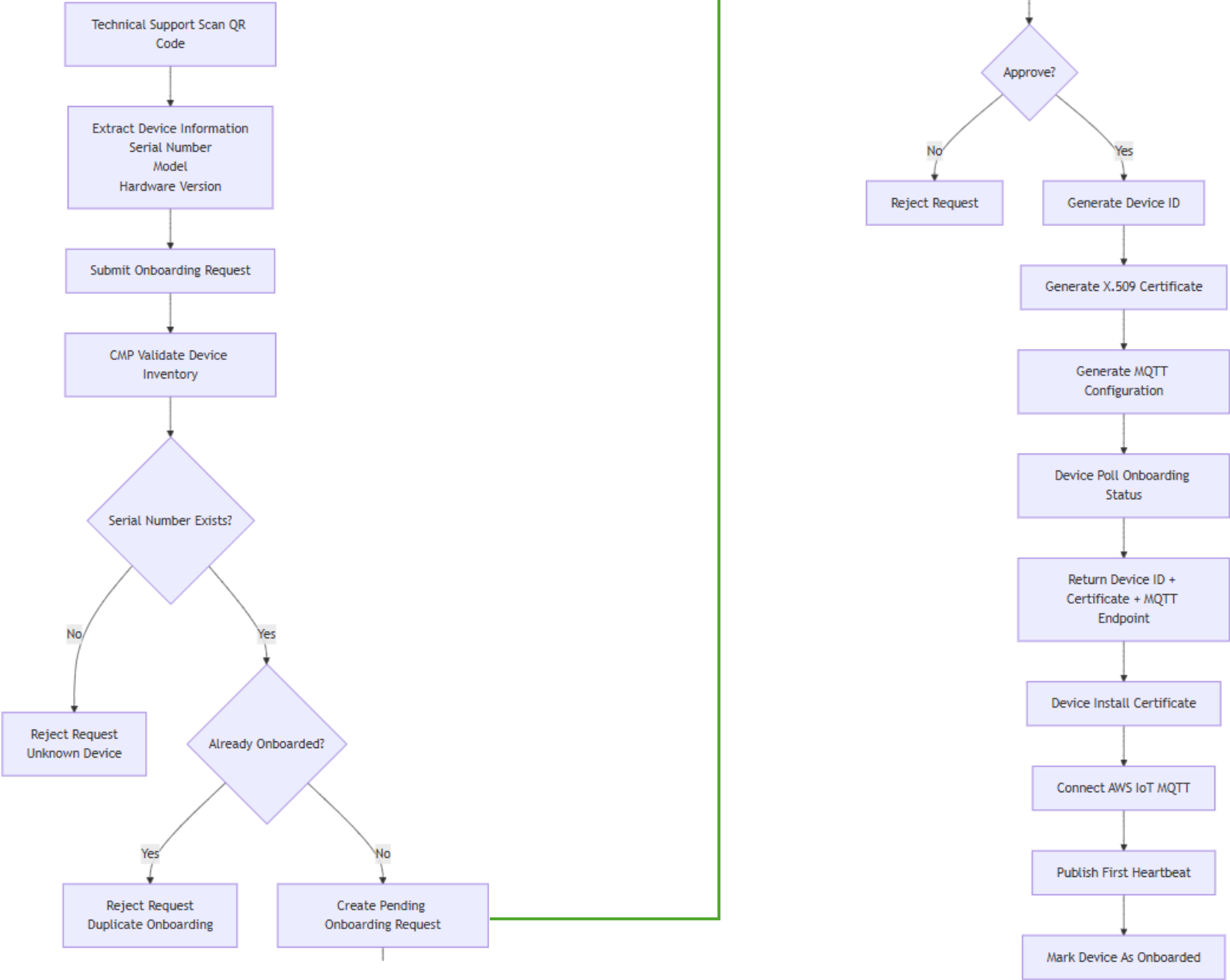
Negative Case 4: Device Failed Certificate Installation



Negative Case 2: Device Already Onboarded



Onboarding Process Flow Chart



Underlying Database Design

Device Side Database Design (Android Edge)